

CMPUT 229

COMPUTER ORGANIZATION AND ARCHITECTURE I

LECTURES DELIVERED BY
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Introduction and Course Outline

- Different structure from previous years (past: 5 or 6 labs, labs are still take home labs but w/o marks, there are now lab exams, the previous online quizzes are now available w/o marks)
- Some lab sessions will be mandatory to attend to complete the lab exam (completed within one hour)
- Code is written in RISC-V Assembly
- The course covers hardware/architecture, math calculation and programming in Assembly
- The midterm now only covers concepts/theory w/o a coding portion
- Hardware and the interaction with it (how computers actually work)
- What is the hardware we interact with?
- High-level chip design, performance characteristics of computers and evaluating performance (at a high level)
- Assembly programming
- Assembly programming is tedious! (but not hard)
- Canvas contains all the set of slides, course discord, former lecture videos (there as a resource but slightly outdated, available for a second perspective)
- The course follows the textbook, use it alongside the lectures

Computer Abstractions and Technology

CHAPTER 1

Reading includes Section 1.1 - 1.5

THEOREM 1.1 (Moore's Law). *The complexity for minimum component costs has increased at a rate of roughly a factor of two per year*

Moore's Law does not typically hold up in modern times due to physical constraints. Nonetheless computers are still pervasive in modern times

There are a variety of **types** of computers

- Desktop computers (made for general purpose)
- Server computers (network-based, high capacity, performance and reliable)
- Embedded computers (hidden as components of systems, constrained by performance/costs/power)

1.1. Below the Program

Application software is written in a high level-language (Java, C++) Below that, is the system software, a compiler that translates HLL code to machine code (as well as the operating system) Below that, is the hardware (I/O, processors, etc.)

All kinds of computers contain similar components, input which travels over memory on a “datapath” to the processor and returned as output

- Hard drive
- Processor
- Memory
- Motherboard

DEFINITION 1.2 (Datapath). *Performs operations on data*

DEFINITION 1.3 (Control). *Sequences datapath, memory, etc.*

DEFINITION 1.4 (Cache memory). *On-board memory, small fast SRAM for immediate access to data*

1.2. The Big Picture

- Abstractions helps us deal with this supposed complexity in a computer
- Learning the lower level concepts make us better computer scientists however
- Instruction set architecture (ISA) is prescribed by the hardware
- Application binary interface is the ISA + System Software Interface (operating system)

1.3. *Manufacturing Integrated Circuits (ICs)*

Integrated Circuits are manufactured using silicon ingots (silicon is used because it is a semiconductor), ingots are cut into blank wafers and processed into a patterned wafer. These patterned wafers are tested and then cut into “dies” (similar to a chip)